

NKEMJIKA GRACE NNAMA

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EDUCATION

Ph. D. in Interdisciplinary Aging Program.

Tulane University School of Medicine, New Orleans, USA

M. Sc. in Anatomy (Neurobehavioral Sciences).

Nnamdi Azikiwe University, Nnewi Campus, Nigeria.

Thesis title: Relationship between Fingerprint patterns, Multiple intelligence, and Emotional intelligence among young adults in Nnamdi Azikiwe University, Nnewi Campus.

B. Sc. in Human Anatomy

University of Port Harcourt, Nigeria.

Relevant Coursework: Neurosciences, Pharmacology, Gross and Neuroanatomy, Advanced Anatomical Techniques, Microscopy and Tissue processing, Genetics.

RESEARCH EXPERIENCE

Laboratory for Experimental and Translational Neurobiology (LETNeu), Ondo state, Nigeria.

2024.

Position: Graduate Research Intern/ Visit.

- Performed immunohistochemistry on brain tissues using antibodies; GFAP, Iba1, Parvalbumin, Alpha synuclein e.t.c.
- Conducted imaging and imaging analysis with the use of Toup blue and Image J.

Faculty of Basic Medical Sciences, Nnamdi Azikiwe University, Nigeria

2022- 2024.

Position: Research Assistant.

Project Topic 1: Ameliorative Effects of Palm Oil on Sniper Induced Toxicity on the Cerebral Cortex of Adult Wistar Rats.

- Extracted the brain tissue, fixed, and passed it through routine histological analysis such as dehydration, clearing, impregnation and embedding, sectioning, staining, and microscopy.
- I investigated the neurotoxic effect of 2,2- dichlorovinyl dimethyl phosphate on the cerebral cortex which revealed a significant increase in proliferation of glial cells and neuronal damage.

Project Topic 2: Effects of Aqueous extract of Calyx of *Hibiscus sabdariffa* on the Testis of Alloxan Induced Diabetic Adult Wistar Rats.

- Extracted testicular tissues, fixed, and conducted histological analysis.
- Conducted analysis of sperm morphology, motility, and sperm count using ANOVA.
- Prepared manuscript for publication.

Project Topic 3: Investigating the Effects of Methamphetamine on the Amygdala of Adult Male Wistar Rats.

- Extracted the brain tissue and performed biochemical analysis for oxidative stress.
- Performed routine histological procedures on brain tissues to investigate the histological alterations and neurotoxic effects of methamphetamine on the amygdala.
- Prepared manuscript for publication.

Position: Graduate student.

- Investigated the relationship between Fingerprint Patterns, Multiple Intelligence and Emotional Intelligence among young adults.
- Recruited about 300 human participants, administered questionnaires, and collected data.
- Conducted data analysis using SPSS.

FUNDED TRAININGS/ WORKSHOPS

Mind over Adversity: Exploring the Neurochemical Dynamics of Post Traumatic Stress Disorders, University of Rwanda, Kigali. Funded by International Society for Neurochemistry (ISN). **2025.**

- Neurotransmitters in health and diseases.
- The use of open- source cloud tools in PTSD.
- Statistical analysis.

Gene- Environment Interactions in Neurological Disorders: A Contemporary Update, Laboratory for Experimental and Translational Neurobiology (LETNeu), UNIMED, Ondo State, Nigeria. Funded by the International Society for Neurochemistry (ISN). **2024.**

- Immunohistochemistry skills.
- Performed synchronization using *C. elegans* experimental model and behavioral assays
- Metal transporters such as DMT-1 in neurological diseases.
- *In vivo* fluorescent imaging, microscopic imaging and analysis.

Neuro-Upscale Training Workshop, LAUTECH Neuroscience Group, Oyo State, Nigeria.

Funded by the International Brain Research Organization (IBRO). **2024.**

- Molecular biology skills (Performed DNA/ RNA isolation, Introduced to PCR, Gel electrophoresis and Gene sequencing).
- Genomics and Bioinformatics.
- Introduced to Computational Neuroscience.

Workshop on Stereotactic Neurosurgery with Laboratory Rats, Rivers State University, Nigeria. Funded by Training and Research in Natural Sciences for Development (TReND). **2022.**

- I performed stereotactic surgeries in rodents with the stereotax machine and craniotomy.
- I determined brain co-ordinates with the use of brain atlas, also performed electrophysiology in rodents.
- Introduced to Optogenetics.

School of Neurogenetics and Scientific Survival Skills for Young Neuroscientists, Abuja, Nigeria. Funded by International Brain Research Organization- Africa Regional Committee (IBRO- ARC). **2022.**

- CRISPR/ Cas 9 gene editing, Pharmacogenomics and its application.
- Laboratory Mathematics and Data Analysis.
- Introduction to Grant writing.

RESEARCH SKILLS AND TECHNICAL COMPETENCY

- Neurobehavioral analysis/ tests.
- Immunohistochemistry and imaging analysis.
- Synchronization using the *C. elegans* experimental model; behavioral tests such as ethanol preference, locomotion assay, basal slowing response.
- Microscopy (Bright field, fluorescent and light microscopes)
- Rat brain dissection, craniotomy and electrophysiology.
- Histologic tissue preparation and processing (Hematoxylin and Eosin, Cresyl violet): Tissue excision.
- Survival and aseptic surgery in rodents.

LIST OF PUBLICATIONS

- Ezejindu D. N., Ugwuja J. O., **Nnama N. G.**, Okafor E. C., Enemuo I. C. (2024). The Effect of Aqueous Extract of Calyx of *Hibiscus sabdariffa* on the Testis of Alloxan Induced Diabetic Adult Wistar Rats. *Journal of Anatomical Sciences*. 15(1); 49-55.
- **Nnama N. G.**, Ezejindu D. N., Udodi P. S., Agulanna A. E., Enemuo I. C., Okafor E. C. (2023). Ameliorative Effects of Palm Oil on Sniper Induced Toxicity on the Cerebral Cortex of Adult Male Wistar Rats. *World Journal of Advanced Research and Reviews*. 19(2); 809-815. <https://doi.org/10.30574/wjarr.2023.19.2.1496>
- **Nnama N. G.** and Ezejindu D. (2021). Relationship between fingerprints patterns and Multiple Intelligence among young adults in Nnamdi Azikiwe University, Anambra State. *World Journal of Advanced Research and Reviews*. 12(1); 349-354. <https://doi.org/10.30574/wjarr.2021.12.1.0445>
- 'Fetal Development (Part 1) for Student Nurses with Multiple Choice Questions and Answers' by **Nnama N. G.** (E-Book, 2024).

REFERENCES

Available on request.